

JUNEAU, AK, USA

WMO: 703810

Lat: 58.357N

Lon: 134.564W

Elev: 16

StdP: 14.69

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 25309

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.6	9.8	-10.7	3.1	11.0	-5.0	4.2	14.9	31.3	38.9	27.6	37.1	6.3	90	0.685

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	74.4	59.7	70.5	58.4	67.0	56.7	61.3	71.9	59.6	67.7	58.2	64.5	8.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.3	70.1	61.7	56.2	67.4	60.8	55.1	64.7	59.8	27.2	71.8	26.1	67.8	25.1	64.3	65.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
26.2	23.0	19.4	DB	0.8	80.2	4.5	3.3	-2.4	82.6	-5.1	84.5	-7.6	86.3	-10.8	88.7
			WB	-0.1	63.8	4.3	1.7	-3.2	65.0	-5.6	66.0	-8.0	66.9	-11.1	68.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.3	28.8	30.0	33.0	40.9	49.1	54.8	57.1	56.3	50.3	42.4	33.7	30.2	
	DBStd	12.19	9.27	7.97	7.00	4.66	5.09	4.54	3.83	3.81	3.52	4.62	7.39	8.65	
	HDD50	3485	657	560	528	275	79	6	1	1	37	237	489	615	
	HDD65	8295	1122	980	993	723	492	308	246	271	440	701	939	1080	
	CDD50	668	0	0	0	3	51	149	222	195	47	1	0	0	
	CDD65	5	0	0	0	0	0	2	2	1	0	0	0	0	
	CDH74	118	0	0	0	0	3	46	45	24	0	0	0	0	
	CDH80	10	0	0	0	0	0	5	4	1	0	0	0	0	
Wind	WSAvg	7.2	7.0	7.0	7.7	7.6	7.3	6.7	6.4	6.6	7.5	7.9	7.3	7.5	
Precipitation	PrecAvg	57.2	4.7	3.9	3.5	2.8	3.4	3.1	4.5	5.5	7.5	7.9	5.3	5.2	
	PrecMax	85.2	9.1	8.5	6.4	7.5	9.2	6.0	10.4	12.3	15.2	15.2	11.1	13.6	
	PrecMin	41.1	0.9	0.1	0.6	0.5	0.8	1.1	1.1	0.6	2.3	3.5	1.1	0.5	
	PrecStd	10.7	2.0	2.2	1.5	1.4	1.6	1.1	1.9	2.4	2.8	2.6	2.4	2.6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.1	45.9	51.4	63.8	73.0	79.8	79.2	77.2	67.3	55.0	49.4	47.6	
		MCWB	43.3	40.4	44.5	49.3	54.9	60.7	62.0	61.5	54.6	50.7	46.4	44.7	
	2%	DB	43.6	43.2	46.3	57.3	68.4	73.2	73.7	72.6	62.4	52.2	46.4	44.5	
		MCWB	41.8	40.7	40.8	45.2	53.7	58.7	61.1	60.0	53.4	48.7	44.2	42.7	
	5%	DB	41.3	41.3	43.6	53.2	64.0	68.4	69.4	68.3	58.9	50.3	44.5	42.3	
		MCWB	39.8	39.4	39.7	44.0	51.5	57.1	59.4	58.7	53.2	47.5	42.5	40.8	
	10%	DB	39.3	39.5	41.6	49.7	60.0	64.4	65.1	64.3	56.4	49.0	42.4	40.0	
		MCWB	38.2	37.7	38.7	42.5	49.8	55.2	57.5	57.3	52.7	46.6	40.8	38.7	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.4	42.7	45.0	50.6	57.2	61.9	63.7	63.3	58.2	52.3	46.7	45.6
			MCDB	45.8	43.8	48.2	61.4	69.4	75.7	75.7	74.2	61.3	53.9	48.2	46.4
2%		WB	41.9	41.1	42.4	47.2	54.8	59.6	61.6	61.1	56.1	50.1	44.5	43.1	
		MCDB	43.2	42.6	45.3	54.9	65.6	71.2	72.4	70.1	59.2	51.4	46.1	44.4	
5%		WB	39.9	39.6	40.7	45.3	52.7	57.8	59.6	59.5	54.5	48.4	43.0	41.0	
		MCDB	41.1	41.1	42.9	50.9	61.6	66.6	67.4	66.2	57.3	49.7	44.4	42.3	
10%		WB	38.3	38.1	39.1	43.7	51.1	55.9	58.1	58.0	53.2	47.0	41.1	38.8	
		MCDB	39.3	39.5	40.9	48.5	58.1	62.9	64.1	62.9	55.7	48.5	42.2	39.8	
Mean Daily Temperature Range	5% DB	MDBR	7.7	9.8	11.0	14.3	15.5	13.9	12.3	12.3	10.5	8.8	7.8	7.5	
		MCDBR	7.2	9.9	14.2	22.8	26.0	24.2	23.6	23.0	17.9	11.0	7.5	7.2	
		MCWBR	5.7	7.4	9.6	12.5	12.7	11.9	12.3	12.4	10.6	7.8	5.8	6.0	
	5% WB	MCDBR	6.8	8.2	11.4	18.2	21.5	21.3	19.6	18.8	12.8	8.1	6.7	7.0	
		MCWBR	5.8	6.6	8.4	10.8	11.0	10.8	11.0	10.9	8.5	6.6	5.7	6.0	
Clear-Sky Solar Irradiance	taub	0.292	0.353	0.329	0.394	0.412	0.425	0.429	0.390	0.374	0.330	0.287	0.266		
	taud	2.079	1.997	2.155	2.000	2.072	2.151	2.198	2.347	2.322	2.310	2.127	2.045		
	Ebn at Noon	161	198	255	253	258	257	253	256	239	215	161	129		
	Edh at Noon	17	30	34	47	47	44	41	33	29	22	16	12		
All-Sky Solar Radiation	RadAvg	120	350	733	1224	1564	1564	1334	1093	695	365	162	82		
	RadStd	15	65	100	141	175	116	148	141	115	56	26	10		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (13 neighbors)	N/A	+2.11	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page